

Reachability Search over Convex Sets for Multi-Regrasp Manipulation Planning

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[TODO: emails]

Abstract—Many manipulation tasks need a sequence of grasps whose length is not known in advance. Turning a cap with a wrist of limited range takes as many regrasps as the goal, the kinematics and the obstacles require. We cast multi-regrasp planning as a reachability problem over a convex decomposition of the collision-free configuration space of robot and object. Grasp pieces, the polytopes where the robot can hold the object, are the only sets where the object moves. Inspired by recent footstep planners, we search them with an A* whose nodes are not configurations but polytopes paired with the set of object configurations reachable inside them. When one object coordinate is coupled to one robot joint, this set is an interval, so reachability propagates by interval arithmetic and dominance is exact. The graph is built with linear programs only, and the search returns a plan with the minimum number of grasp actions relative to the decomposition, with no integer variables and no fixed horizon. On a 10-DOF cap-turning task with a Franka Panda, where a mixed-integer formulation is numerically unreliable, it plans in 4 ms. The same task shows that probabilistically certified decompositions can contain collisions concentrated on the grasp sets where regrasps happen. This makes the fidelity of these switch sets, not coverage of free space, the key requirement for planners of this kind. [TODO: headline numbers from E1–E4.]

Index Terms—Manipulation planning, regrasping, reachability, graph search, convex decompositions.

I. INTRODUCTION

Many manipulation tasks cannot be completed with a single grasp. Turning a bottle cap with a wrist of limited range, cranking a valve, or reorienting an object the hand cannot rotate far enough all force the robot to grasp, move, release, reposition and grasp again. How many regrasps are needed, and where, depends on the goal, on the robot’s kinematics and on the obstacles around the object. In our 10-DOF task, turning a cap by 4rad with a wrist that sweeps 2rad takes at least two grasps in free space. Obstacles that block part of the sweep raise this to five (Section VI-C). The question to answer is one of *reachability*: which object configurations can be reached with k grasps, and from which robot configurations each grasp can start and end.

Footstep planning has the same structure and has made it explicit. A footstep plan is a sequence of contacts of unknown length, each constrained by where the previous ones allow the next foot to land. Planners moved from A* over discretised footholds [1], [2] to mixed-integer programs over convex terrain decompositions [3], [4], made fast by ℓ_1 relaxations [5],

[6], and to planners built on explicit reachability conditions [7]. Most recently, CASSR [8] runs A* over nodes that are convex sets of footholds reachable from the start. It plans 30 footsteps in under 125 ms, outperforming discretised A* by up to two orders of magnitude, and a commercial mixed-integer solver. In footstep planning, *the number of contacts is an output of a reachability computation over convex sets*.

Prehensile manipulation shares this combinatorial core, yet long regrasp sequences have received much less attention. Siméon et al. [9] formalised manipulation planning in the composite configuration space of robot and object. Paths alternate *transit*, where the robot moves alone, and *transfer*, where it holds the object, and switch where grasp and placement sets intersect. They explored the resulting manipulation graph with probabilistic roadmaps. Since then, regrasp sequences have mostly been planned over finite sets of grasps and placements [10]–[12]. With continuous grasp sets, planners sample, which gives at best probabilistic completeness and is slow in the thin sets where grasps happen [9], [13], [14], or optimise over a skeleton whose number of mode switches is guessed or enumerated [15], [16]. To our knowledge, no manipulation planner offers what CASSR offers for footsteps: a search whose cost is the number of contact switches, whose nodes are continuous reachable sets, and which is complete and optimal relative to its model of free space. [CHECK: novelty; closest are [16], [17].]

Learning does not close this gap, and multi-regrasp tasks show why. Learned policies have reached impressive dexterity, from in-hand reorientation [18] to twisting lids off with two multi-fingered hands [19], and imitation learning covers many everyday tasks [20]. But the minimum number of regrasps is a piecewise-constant function of the goal and the scene, roughly $\lceil D/L \rceil$ for a turning angle D and a longest collision-free wrist sweep L . A policy trained on goals that need k regrasps must extrapolate to $k + 1$, a longer horizon than any it was trained on, where errors compound [21]. Moving an obstacle slightly can change L and with it the discrete structure of the solution, not only its geometry. A policy also cannot certify that a goal is unreachable with fewer grasps. A planner that computes reachability on an explicit model handles these cases by construction: a new goal is a new query, and a new obstacle layout is a new model. [CHECK: an argument, not an experiment; keep claims at this level unless a learned baseline is added.]

We bring the reachability view of footstep planning to multi-

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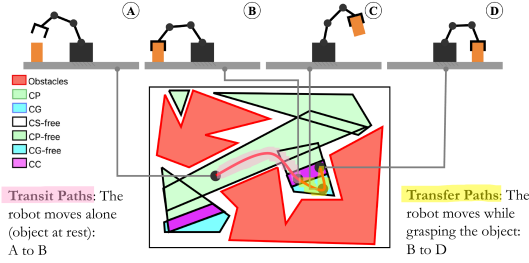


Fig. 1. Transit (A→B) moves the robot alone while the object rests in the placement space C_P . Transfer (B→D) moves it while holding the object, in the grasp space C_G . Plans switch only where the two sets intersect (purple), inside a convex decomposition of free space. [TODO: replace with the Franka cap task next to the abstract picture; relabel CC as $C_G \cap C_P$.]

regrasp manipulation. We build on convex decompositions of the collision-free configuration space [22], [23], the representation behind globally optimal motion planning with Graphs of Convex Sets [24]. We intersect them with polytopes of grasping configurations. The resulting *grasp pieces* are the only sets in which the object moves, the switch sets of the manipulation graph. Our A* pairs each polytope with the set of object configurations reachable inside it. When one object coordinate is coupled to one robot joint, this set is an interval, propagation is interval arithmetic and dominance is exact. We arrived at this design after a mixed-integer formulation [25], which solved low-dimensional tasks, became numerically unreliable on a 7-DOF arm. Its grasp polytopes are 10^{-3} rad thick, nine orders of magnitude below its big-M constant. [CHECK: anonymise the self-citation if double-blind.] Our contributions are:

- a formulation of multi-regrasp planning as reachability over a convex decomposition, with grasp pieces as the switch sets between transit and transfer (Sections III, IV-C);
- a reachability A* with an LP-only graph, an admissible and consistent heuristic and exact dominance. It returns a plan with the minimum number of grasp actions relative to the decomposition, of a length not fixed in advance, and a solver-free stage turns that plan into certified straight motions (Sections IV-D–IV-F);
- grasp polytopes for a 7-DOF arm, built from a linearisation at an NLP sampling seed [26] that uses the exact invariance of the grasp to wrist rotation and a trust region (Section IV-B);
- an evaluation up to the 10-DOF space of a Franka arm, hand and cap, with an analysis of the MIQP’s failure. It shows that probabilistically certified decompositions concentrate their errors on grasp sets (Sections V–VII). [TODO: update with E1–E4.]

All guarantees are relative to the decomposition and grasp polytopes, and are only as meaningful as these sets are faithful near contact, which is where our experiments expose a weakness.

II. RELATED WORK

A. Manipulation and regrasp planning

Regrasp planners first searched over finite sets of grasps and placements [10], [11], an approach that continues in regrasp graphs over thousands of precomputed grasps [12]. Lertkultanon and Pham [27] certify completeness for bimanual regrasping relative to such a discretisation. The manipulation graph [9] made grasps and placements continuous, and was generalised to multi-modal planning [13], [28], asymptotically optimal sampling [29], [30] and constraint graphs with constrained samplers [14], [31], [32]. TAMP adds symbolic decisions on top [33], [34]. In optimisation-based TAMP, an outer search enumerates skeletons with a fixed number of mode switches [15], and regrasp maps [16] guide that search towards regions where several regrasps are possible. Closest in spirit to our work, Liu et al. [17] precompute a graph of mutually reachable sets of object orientations and grasps for planar contact-rich manipulation. We also reason about reachable sets. Ours are polytopes of the robot–object configuration space, and we guarantee completeness and the minimum number of grasps relative to the decomposition.

B. Reachability in contact planning

Our search follows CASSR [8]. There, a node is the subset of a contact surface reachable from its parent. It is expanded by Minkowski sums and intersections with neighbouring surfaces, and a quadratic program places the footsteps once the surfaces are fixed. Regrasping changes two things. Reachability grows only inside grasp pieces, and with one coupled object coordinate the reachable sets are intervals. Dominance is therefore exact and the heuristic admissible, whereas CASSR scales its heuristic, which then is not admissible, and discretises yaw.

C. Convex decompositions and Graphs of Convex Sets

Configuration-space obstacles cannot be described exactly in general [35], [36], so free space is inner-approximated by convex sets. IRIS grows such sets by alternating separating hyperplanes and inscribed ellipsoids [37]. C-IRIS certifies them [38], [39], while IRIS-NP, IRIS-NP2 and IRIS-ZO trade certification for speed and give probabilistic guarantees [23], [40]. Clique covers automate the seeding [22], and GPU implementations approach online rates [41]. Graphs of Convex Sets (GCS) find optimal paths over decompositions through a tight convex relaxation [24], [42]. Extensions cover non-Euclidean joints [43], temporal logic [44] and planar contact-rich manipulation [45], and heuristic searches handle graphs too large to solve at once [46], [47]. GCS* prunes partial paths by domination checks on reachable sets, which in general must be approximated [46]. With our reachable sets, the check is exact.

III. PROBLEM FORMULATION

A. Transit, transfer and switch sets

A robot with configuration $x \in \mathcal{X} \subset \mathbb{R}^{n_r}$ (arm and fingers) manipulates an object with configuration α in a known

static scene. The state $q = (x, \alpha)$ lives in the composite configuration space $\mathcal{C}$ [9], of dimension $n = n_r + n_o$, and $\mathcal{C}_{\text{free}} \subseteq \mathcal{C}$ is the set of states without collisions other than the contacts a grasp requires. The object cannot move on its own. In the *placement space* $\mathcal{C}_P$ it rests stably, and in the *grasp space* $\mathcal{C}_G$ the robot holds it. A manipulation path alternates *transit* segments in $\mathcal{C}_P$, where the robot moves alone and α is constant, and *transfer* segments in $\mathcal{C}_G$, where the robot carries the object with a fixed relative pose. Segments switch on $\mathcal{C}_G \cap \mathcal{C}_P$, where the robot can grasp or release the object.

B. Multi-regrasp tasks with a coupled object coordinate

We study the class of tasks captured by the following assumption, which includes caps, knobs, valves and other objects that rotate about a fixed axis.

Assumption 1: (i) The object has one coordinate $\alpha \in [\alpha_{\min}, \alpha_{\max}]$ and is always stably placed, so $\mathcal{C}_P = \mathcal{C}$. (ii) Holding the object couples α to one robot joint w , the *wrist*: during a transfer, $\dot{\alpha} = -\dot{w}$. The sign follows from the orientation of the axes. (iii) Neither the grasp constraints nor collisions depend on α : $\mathcal{C}_G = G \times [\alpha_{\min}, \alpha_{\max}]$ and $\mathcal{C}_{\text{free}} = \mathcal{X}_{\text{free}} \times [\alpha_{\min}, \alpha_{\max}]$ for some $G, \mathcal{X}_{\text{free}} \subseteq \mathcal{X}$. Point (iii) holds for rotationally symmetric objects, such as the cap in our experiments, whose collision geometry is a cylinder.

Since w has a limited range $[w_{\min}, w_{\max}]$, turning the object by more than the range requires several grasps. Without loss of generality, let $\alpha_{\text{goal}} \leq \alpha_{\text{init}}$. Since α affects neither collisions nor grasps, a plan never gains from turning the object backwards. It therefore works like a ratchet. A *stroke* grasps the object and moves the wrist forward, so α decreases by the wrist travel. A *return* releases the object, rotates the wrist back with α fixed, and grasps again. A *release* lets go of the object before the robot leaves the grasp set. Free motion between grasps is transit in $\mathcal{X}_{\text{free}}$, with α fixed.

C. Problem statement

Given $q_{\text{init}} = (x_{\text{init}}, \alpha_{\text{init}})$ and $q_{\text{goal}} = (x_{\text{goal}}, \alpha_{\text{goal}})$, find a continuous path from q_{init} to q_{goal} in $\mathcal{C}_{\text{free}}$, made of transit segments and strokes, that minimises the number of grasp and ungrasp actions (strokes, returns and releases). Ties are broken by the number of changes of convex set along the path, and, secondarily, by path length.

In the reachability view, let $\mathcal{R}_k \subseteq \mathcal{C}_{\text{free}}$ be the set of states reachable from q_{init} with at most k grasp actions. The minimum number of actions is the smallest k such that $q_{\text{goal}} \in \mathcal{R}_k$. Neither $\mathcal{R}_k$ nor $\mathcal{C}_{\text{free}}$ can be computed exactly. Our planner computes $\mathcal{R}_k$ exactly for an inner approximation of $\mathcal{C}_{\text{free}}$ and $\mathcal{C}_G$ by convex polytopes, and its guarantees hold relative to that approximation.

IV. METHOD

The planner has an offline stage, which depends only on the robot, the object and the scene, and an online stage per query. Offline, it decomposes free space into polytopes (Section IV-A), builds grasp polytopes (Section IV-B), and intersects them into grasp pieces and a polytope graph using

only LPs (Section IV-C). Online, a reachability A* chooses the sequence of sets and grasp actions (Section IV-D), and a placement stage turns it into certified motions (Section IV-E). Fig. 2 illustrates both stages on our 10-DOF task.

A. Convex decomposition of free space

We inner-approximate $\mathcal{C}_{\text{free}}$ by polytopes $P_j = \{q : A_j q \leq b_j\}$, $j = 1, \dots, J$, computed with clique covers of visibility graphs [22] and IRIS-NP2 regions [23] in Drake [48]. Each region is collision-free up to a fraction ε of its volume, with confidence $1 - \delta$. Certified alternatives exist [39] but scale poorly to 10 dimensions. By Assumption 1(iii), the object coordinate does not affect collisions. We therefore replace each region by its slice at $\alpha = 0$, $F_j = \{x : (x, 0) \in P_j\}$, dropping rows that only bound α . This is sound: if $(x, 0)$ is collision-free, so is (x, α) for every α . The robot can then occupy any $x \in F_j$ at any object angle, and the planner tracks α separately.

B. Grasp polytopes for a 7-DOF arm

For a gripper with a few joints, G is a box in configuration space [25]. For a 7-DOF arm, a grasp constrains the pose of the hand frame E relative to the cap frame C , a nonlinear function of the arm joints. The hand must be centred above the cap, within a height band, with its approach axis $z_E = {}^C R^E(x) e_z$ pointing down:

$$h(x) = [p_x, p_y, z_{E,x}, z_{E,y}]^\top = 0, \quad (1)$$

$$g(x) = [p_z - z_{\max}, z_{\min} - p_z, z_{E,z} + c]^\top \leq 0, \quad (2)$$

with $p = {}^C p^E(x)$, $[z_{\min}, z_{\max}] = [0.105, 0.110]$ m and $c = 0.9$. The fingers are restricted to a 1 mm band around the cap width. The set $\{h = 0, g \leq 0\}$ is a thin, curved manifold. We find a seed grasp x_0 on it with the Gauss–Newton phase of NLP sampling [26] and linearise around it:

$$G = \{x : |h_0 + J_h(x - x_0)| \leq \epsilon, \quad g_0 + J_g(x - x_0) \leq \epsilon, \\ x_{\min}^G \leq x \leq x_{\max}^G, \quad |x_j - x_{0,j}| \leq \rho \quad \forall j \in \mathcal{A}\}, \quad (3)$$

where h_0, J_h, g_0, J_g are evaluated at x_0 , $\epsilon = 10^{-3}$, and $\mathcal{A}$ are the arm joints other than the wrist. Two properties of the task shape this construction. First, the wrist axis passes through the origin of E along z_E , so rotating the wrist changes neither p nor z_E : h and g are *exactly* invariant along w . We verified this to machine precision over the whole wrist range. The wrist columns of J_h and J_g therefore vanish, and G spans the full wrist range. Second, J_h has rank 4, so without the last constraint in (3) the null directions of J_h would be bounded only by the joint limits. The extreme points that the graph construction computes would then lie where the linearisation is meaningless. Without this *trust region*, they moved an arm joint by 2.27 rad from x_0 , with $\|h\|_\infty = 0.95$ and the hand 0.80 m away from the cap. With $\rho = 0.05$ rad, $\|h\|_\infty$ stays around 2×10^{-3} along our plans [CHECK: guide numbers, q0 run]. Seeds drawn from other components of the grasp manifold, e.g., with NLP sampling’s manifold-RRT interior phase, give further grasp polytopes G^s that the search treats in the same way. We use a single seed in this paper.

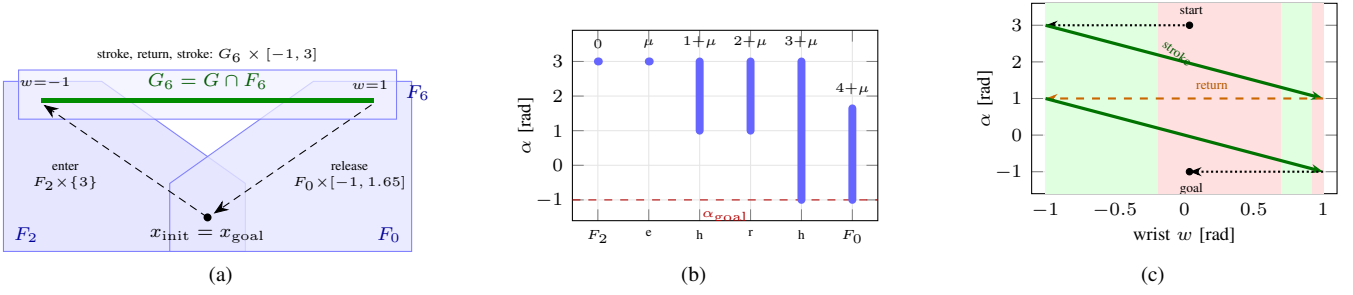


Fig. 2. Reachability A* on the 10-DOF cap task, turning the cap from $\alpha_{\text{init}} = 3$ to $\alpha_{\text{goal}} = -1$ rad with the arm returning to its start configuration. (a) Sets on the returned path. The arm enters the grasp piece G_6 from region F_2 , performs stroke–return–stroke, and releases into F_0 . (b) The interval $[\alpha^-, \alpha^+]$ of each node on the path (bars; e, h, r: G_6 entered, holding, released), labelled with its cost-to-come g . Only strokes widen the interval, and the release lowers α^+ by the forced turning $d = 1.35$ rad. (c) Placed waypoints projected on the wrist–cap plane: strokes turn the cap ($\Delta\alpha = -\Delta w$), and returns and transit keep it fixed. Shading shows the true collision status of the wrist sweep at the seed grasp (green: free, red: in collision). Every region contains the full sweep, so the plan is valid relative to the decomposition but not physically (Section VI-C).

C. Grasp pieces and the polytope graph

For every region with $F_k \cap G \neq \emptyset$ we form the *grasp piece* $G_k = G \cap F_k$. Grasp pieces are the switch sets of the manipulation graph, and the only sets in which the object moves. Two LPs give the extreme wrist values $w_k^\pm = \max/\min\{\pm w : x \in G_k\}$ and points $x_k^\pm \in G_k$ attaining them. A stroke inside G_k turns the object by at most $L_k = w_k^+ - w_k^-$, and we discard pieces with $L_k < L_{\min}$. Since G is thin in every direction but the wrist, a piece is essentially a wrist interval, and by convexity the segment $[x_k^-, x_k^+]$ lies in G_k .

The graph has two kinds of edges. A *transit* edge joins F_i and F_j if they intersect. Its door point is the Chebyshev centre of $F_i \cap F_j$. A *switch* edge joins F_i and G_k if they intersect. For it, we store the wrist range $[a_{ik}^-, a_{ik}^+]$ of $F_i \cap G_k$ and the points attaining its ends. Every quantity is an LP feasibility check, a Chebyshev centre or an extreme point. No binaries and no vertex enumeration are needed, and the number of LPs is $O(J^2 + JK)$ for J regions and K pieces.

D. Reachability A*

a) *Nodes*: Following CASSR [8], a node does not represent a configuration but a set reachable from the start. A *transit node* $(F_i, [\alpha^-, \alpha^+])$ states that every $x \in F_i$ is reachable together with every $\alpha \in [\alpha^-, \alpha^+]$. Since F_i is convex and α is frozen during transit, any point of F_i can be reached from any other along a straight line. A *grasp node* $(G_k, [\alpha^-, \alpha^+], \phi, [a^-, a^+])$ additionally records the wrist range $[a^-, a^+]$ through which the piece was entered and a phase ϕ . In phase *entered*, the robot is in the piece and has not grasped yet. In phase *holding*, it has just completed a stroke. In phase *released*, it has completed a return and the wrist is back at w_k^- . All path information is carried by the interval. Reaching F_3 before any grasp and after a stroke of 2 rad gives the same polytope with intervals $\{3\}$ and $[1, 3]$, two different nodes.

b) *Moves*: Table I lists the successors of a node. Only strokes widen the interval: a stroke can turn the object by any amount up to its cap, so it lowers α^- and keeps α^+ . Since grasping happens only inside grasp pieces, these are the only sets where reachability grows in α , which is the

TABLE I
MOVES OF THE REACHABILITY A*. $c_1 = w_k^+ - a^-$ AFTER ENTERING, $c_1 = L_k$ AFTER A RETURN; $d = \max(0, a_{jk}^- - a^+)$.

Move	From $\rightarrow$ to	Interval update	Cost
transit	$F_i \rightarrow F_j, F_i \cap F_j \neq \emptyset$	–	μ
enter	$F_i \rightarrow G_k$ (entered), $F_i \cap G_k \neq \emptyset$	–	μ
stroke	entered/released $\rightarrow$ holding	$\alpha^- \leftarrow \alpha^- - c_1$	1
return	entered/holding $\rightarrow$ released	–	1
release	holding $\rightarrow F_j, F_j \cap G_k \neq \emptyset$	$\alpha^+ \leftarrow \alpha^+ - d$	1
leave	entered $\rightarrow F_j$, only if $d = 0$	–	μ

reachability counterpart of the switch sets of the manipulation graph. Leaving a piece towards F_j can also force turning. Inside G_k the wrist moves forward only while grasped, so reaching the exit range from the entry range turns the object by at least $d = \max(0, a_{jk}^- - a^+)$, which lowers α^+ . We clip α^- at $\max(\alpha_{\text{goal}}, \alpha_{\min})$. Each grasp, return and release costs 1, and each change of set costs $\mu \ll 1$ ($\mu = 10^{-3}$), so the search minimises grasp actions first and set changes second. A release must cost 1, otherwise leaving and re-entering a piece would be a free substitute for a return.

c) *Heuristic, goal and dominance*: With $L_{\max} = \max_k L_k$, the heuristic

$$h(v) = 2 \left\lceil \frac{\alpha^- - \alpha_{\text{goal}}}{L_{\max}} \right\rceil + 1[\phi = \text{holding}] \quad (4)$$

is admissible. Each remaining stroke turns the object by at most $L_{\max}$ and must be followed by an ungrasp, either a return or the final release, and a holding node must still release. It is also consistent, since no move lowers h by more than its cost. A transit node is a goal if $x_{\text{goal}} \in F_i$ and $\alpha_{\text{goal}} \in [\alpha^-, \alpha^+]$. A node is *dominated*, and skipped, if an expanded node with the same key has an interval that contains its interval at no higher cost. The key is the set for transit nodes, and the set, phase and entry range for grasp nodes. Because the abstraction describes reachability completely by key and interval, this check is exact. By contrast, the domination checks of GCS* [46] must in general be approximated, and CASSR uses a non-admissible heuristic [8]. Algorithm 1 summarises the search.

Algorithm 1 Reachability A*

Input: graph of Sec. IV-C; $(x_{\text{init}}, \alpha_{\text{init}}), (x_{\text{goal}}, \alpha_{\text{goal}})$ **Output:** node sequence with grasp actions, or NONE

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1:  $\mathcal{O} \leftarrow \{(F_i, [\alpha_{\text{init}}, \alpha_{\text{init}}]) : x_{\text{init}} \in F_i\}; \mathcal{E} \leftarrow \emptyset$ 
2: while  $\mathcal{O} \neq \emptyset$  do
3:    $\nu \leftarrow \arg \min_{\nu \in \mathcal{O}} g(\nu) + h(\nu)$ ; remove  $\nu$  from  $\mathcal{O}$ 
4:   if  $\nu$  dominated by  $\mathcal{E}$  then continue
5:    $\mathcal{E} \leftarrow \mathcal{E} \cup \{\nu\}$ 
6:   if  $\nu = (F_i, [\alpha^-, \alpha^+]), x_{\text{goal}} \in F_i, \alpha_{\text{goal}} \in [\alpha^-, \alpha^+]$ 
     then
7:     return path to  $\nu$ 
8:   for each successor  $\nu'$  of  $\nu$  (Table I) do
9:     if  $\nu'$  not dominated by  $\mathcal{E}$  then add  $\nu'$  to  $\mathcal{O}$ 
10: return NONE

```

E. Placing configurations

The search fixes the sets and the grasp actions, for example *enter, stroke, return, stroke, release*. As in CASSR, a second stage places the configurations. We do this without a solver. First, a backward pass from α_{goal} allocates the turning to each grasp visit. Visit v turns $T_v = \min(\alpha_v^+, \alpha_v^{\text{after}} + \sum_s c_s) - \alpha_v^{\text{after}}$, where α_v^+ is the upper end of the interval before the visit and c_s are the caps of its strokes. T_v is split among the strokes by water-filling. The pass ends exactly at α_{init} , because the search proved α_{goal} reachable. Second, waypoints are placed: door points for transit, an entry point in $F_i \cap G_k$ at the wrist value from which the first stroke ends at the top of the piece, points on $[x_k^-, x_k^+]$ for strokes and returns, an exit point in $G_k \cap F_j$, and finally q_{goal} . Each waypoint records a *certifying set*, a region F_j or a piece G_k that contains both ends of the straight motion into it. By convexity, it contains the whole motion. We check containment and the coupling rule ($\Delta\alpha = -\Delta w$ on strokes, $\Delta\alpha = 0$ otherwise) for every motion. Path length is not optimised. Since the discrete structure is fixed, a convex program over the chosen sets could shorten the path, e.g., GCS [24] restricted to these sets, or a QP as in CASSR. **[TODO: implement, or keep as future work.]**

F. Properties

Proposition 1 (informal): Relative to the regions $\{F_j\}$ and the pieces $\{G_k\}$: (i) every motion returned is a straight line inside one region or piece that satisfies the coupling of Assumption 1, so the path is collision-free whenever the regions are; (ii) the search terminates, and if a plan exists whose transit motions lie in the regions and whose grasp motions lie in the pieces, it returns one with the minimum number of grasp actions, ties broken by the number of set changes; (iii) placement succeeds on every sequence the search returns.

Sketch. (i) follows from the certifying sets and convexity. (ii) Each stroke lowers α^- by at least $L_{\min}$ until it reaches α_{goal} , so finitely many intervals arise per key and the search terminates. With a consistent heuristic and exact dominance, A* expands a goal node of minimum cost first [49]. Entering

at the highest and leaving at the lowest wrist value of each switch set minimises forced turning, so restricting entries and exits to these points loses no plan. (iii) The backward pass mirrors the interval updates of Table I. **[CHECK: write the full proof; (iii) is asserted in our implementation but not yet proven. A merged chain of overlapping pieces needs chain-aware placement, so each piece is its own segment for now.]** $\square$

The guarantees are relative: a plan the search reports as optimal uses the fewest grasps *that the decomposition allows*. Section VI-C shows that this number can differ from the true one. Path length is not optimal, which is also a stated limitation of CASSR [8].

G. Why not a mixed-integer program?

The same problem can be posed as an MIQP over keyframes $x_0 = q_{\text{init}}, x_1, \dots, x_{2\ell+1} = q_{\text{goal}}$ for a fixed number of grasps ℓ , minimising $\sum_i \|x_{i+1} - x_i\|^2$ subject to the linear transit and transfer constraints. Region membership is enforced with big-M constraints $A_j x_i \leq b_j + M(1 - z_{ij})$, $\sum_j z_{ij} \geq 1$, and binaries for the connected component of each pair. The MIQP is solved for $\ell = 1, 2, \dots$ until feasible [25]. This worked for 3- and 4-DOF tasks, but it is fragile at 10-DOF for three reasons. (a) Big-M: with $M = 10^6$ and MOSEK's default integrality tolerance of 10^{-5} [50], a binary accepted as 1 may relax each row of its region by $M \cdot 10^{-5} = 10$ rad. That is more than any joint range, while grasp polytopes are $2\epsilon = 2 \times 10^{-3}$ rad thick. Tight per-row constants or convex-hull formulations [51] mitigate this at the cost of a larger model. (b) Components: connected components of the switch sets were merged through convex hulls of their vertices, which requires vertex enumeration of thin 10-D polytopes (Qhull fails without joggling). The hull of a union is also not contained in the union. (c) Scaling: the binaries grow as $(2\ell+2)(J+C)$ for C components, and ℓ is found by an outer loop. The reachability A* has none of these issues. It solves LPs offline and does interval arithmetic online, and the number of grasps is an output. **[CHECK: record the MOSEK status/log for the 10-DOF MIQP failure.]**

V. EXPERIMENTAL EVALUATION

Our experiments ask three questions. Does the search find minimum-grasp plans whose length adapts to the query, for new goals, start configurations and obstacle layouts? How does it compare with the MIQP and with sampling-based manipulation planners as the number of required regrasps grows? How faithful are the sets it reasons over?

A. Tasks and implementation

Both tasks turn a cap about a fixed vertical axis with a Franka Panda [52] whose wrist is restricted to $[-1, 1]$ rad, so any turn larger than 2 rad needs several grasps.

- **T1 (3-DOF).** All arm joints but the wrist are locked: $q = (w, x_{\text{finger}}, \alpha)$ with $x_{\text{finger}} \in [-0.045, 0]$ m and $\alpha \in [-\pi, \pi]$. The decomposition has 15 regions. The task turns the cap by 2π .

- **T2 (10-DOF)**. Seven arm joints, two fingers and the cap. The cap is 0.5m in front of the robot base, with two cylindrical obstacles (radius 5mm) next to it. The decomposition is an IRIS-NP2 clique cover with 42 regions [CHECK: coverage target 0.98; (ε, δ) ; generation time]. Grasp polytopes use (3) with $\epsilon = 10^{-3}$, $\rho = 0.05$ rad, and $L_{\min} = 0.05$ rad. The reference query Q0 turns the cap from 3 to -1 rad and returns the arm to its start configuration.

The planner is implemented in Python on Drake [48] (v1.48). The LPs use Drake’s MathematicalProgram, and the MIQP and the GCS segments of the MIQP pipeline use MOSEK [50]. [TODO: hardware; code release.]

B. Protocol

[TODO: Run E1–E4; this is the evidence for the main claim.] **E1, goal sweep:** α_{goal} on a grid over the cap range, with the arm configuration fixed. We report the number of grasp actions against the lower bound $2\lceil D/L_{\max} \rceil$ and the search time. **E2, random queries:** N start and goal arm configurations sampled in the regions, with random turning angles. **E3, obstacle layouts:** M placements of the obstacles, each with its own decomposition. We report offline time (decomposition, grasp pieces, graph) and online time per query. **E4, baselines:** the MIQP of Section IV-G and a sampling-based manipulation planner on the manipulation graph [9], [14], as a function of the number of regrasps the query requires. Metrics are success within a time limit, planning time, grasp actions and path length. For all methods, every returned motion is checked densely with Drake’s collision checker.

VI. RESULTS

A. Reachability A* on the 10-DOF task

Table II and Fig. 2 report query Q0. Of the 42 regions, 8 intersect the grasp polytope. Four of these grasp pieces span the whole wrist range ($L_k = 2$ rad), and the others span between 0.06 and 0.36 rad. The search expands 63 nodes in 4 ms and returns two strokes, one return and one release: enter G_6 from F_2 , stroke, return, stroke, release into F_0 . Its cost, $4 + \mu$, equals the initial heuristic plus one set change, so the heuristic is tight on this query. With $D = 4$ rad and $L_{\max} = 2$ rad, two strokes is the minimum the decomposition allows. The placement stage turns 2 rad per stroke and produces 7 waypoints. All 6 straight-line motions lie in their certifying set and satisfy the coupling rule. The planner never calls a collision checker. Its only geometric operations are the LPs of the offline stage.

B. Comparison with the MIQP

Table III compares the two sequencing methods. On T1, the MIQP returns 11 grasps after proving 10 infeasible. Each stroke turns the cap by 0.604 rad, the wrist range of the grasp set in that decomposition, and $\lceil 2\pi/0.604 \rceil = 11$. GCS then connects all 23 transit and transfer segments. On T2, the MIQP does not return a reliable solution, for the reasons of Section IV-G, whereas the search returns the plan above in

TABLE II
REACHABILITY A* ON T2 (10-DOF), QUERY Q0

Stage	Quantity	Value
Offline	regions J (non-empty after slicing)	42
	grasp pieces K (with $L_k = 2$ rad)	8 (4)
	transit / switch edges	850 / 58
	graph construction time	[TBD]
	decomposition time	[TBD]
Online	nodes expanded	63
	search time	4 ms
	strokes / returns / releases	2 / 1 / 1
	set changes	1
	certified motions	6 / 6

TABLE III
SEQUENCING WITH THE MIQP AND WITH THE REACHABILITY A*

Task	Method	Grasps	Time	Outcome
T1 (3-DOF)	MIQP	11	[TBD]	solved; GCS connected
	Reach. A*	[TBD]	[TBD]	[TBD]
T2 (10-DOF)	MIQP	–	[TBD]	numerical failure [CHECK:]
	Reach. A*	2	4 ms	solved (rel. decomp.)

milliseconds. [CHECK: MIQP outcome and solve time on T2; run the reachability A* on T1 and check it also returns 11.]

C. How faithful are the sets near contact?

A planner that is complete and optimal relative to a decomposition inherits its errors. To measure them where they matter, we swept the wrist over $[-1, 1]$ at the seed grasp, at 99 points, and compared region membership with Drake’s collision checker. The checker finds the wrist free only on $[-1, -0.19]$ and $[0.70, 0.92]$ rad [CHECK: which bodies collide in between: fingers vs. obstacles?]. All four decompositions we computed for T2 (IRIS-NP2 with 42 and 46 regions, and two IRIS-ZO covers with 64 regions [CHECK: the two IRIS-ZO files may be the same cover]) contain every point of the sweep. That gives 53 true positives, 46 false positives and no false negatives in each case. The two-stroke plan of Fig. 2 is therefore valid relative to the decomposition but collides in reality. With the true longest free segment of 0.81 rad, the minimum is $\lceil 4/0.81 \rceil = 5$ strokes, a different discrete structure and not just a different geometry.

This is not a bug in the region generators, but a mismatch between their guarantee and our use. IRIS-NP2 and IRIS-ZO bound the *fraction of volume* of a region in collision [23]. A grasp piece is almost a one-dimensional set of near-zero volume, so it can be entirely in collision without violating that bound. Grasps also happen exactly where the hand is closest to the obstacles. The same sensitivity appeared in low-dimensional versions of the task [25]. With a 4-DOF gripper, a decomposition at 0.70 coverage contained no grasp–release sequence, and at 0.90 it contained one that GCS could not connect. Coverage of free space is the wrong measure for regrasp planning. What matters is fidelity on the switch sets.

(left) E1: grasp actions vs. turning angle D , a staircase tracking $2[D/L]$;
(right) E4: planning time vs. required regrasps for reachability A*, MIQP
and the sampling-based baseline

Fig. 3. [TODO: E1 and E4 results.]

[TODO: E5: re-run Q0 with pieces restricted to checker-validated wrist intervals; expected 5 strokes with no code change.]

D. Generalisation across goals, starts and scenes

[TODO: Results of E1–E4 (Section V-B).]

VII. DISCUSSION AND LIMITATIONS

a) Certify where it matters: The weakest link is not the search but the sets it searches over (Section VI-C). Grasp pieces are thin and almost one-dimensional, so they are much cheaper to validate than full regions. They can be swept densely with a collision checker, split into validated wrist intervals, or certified with C-IRIS [39], and regions can be seeded on grasp sets so that the decomposition is dense where regrasps happen. The search consumes only pieces and their wrist ranges, so it does not change. [TODO: implement piece validation before submission; without it, the 10-DOF plan is not physically valid and a reviewer will say so.]

b) Scope: Exact intervals rely on Assumption 1, which covers caps, knobs, valves and cranks but not general pick-and-place. With more object coordinates, a node’s reachable object configurations become sets that could be propagated with Minkowski sums and intersections as in CASSR [8], at the cost of exact dominance and with nonlinear grasp transforms to approximate.

c) Optimality: The search minimises grasp actions, and then set changes, a weak proxy for length. As in CASSR, the sets are fixed before configurations are placed, so path length is not optimal. A convex refinement over the chosen sets would make each path locally optimal for its sequence, not globally. The heuristic uses the longest piece in the map. A heuristic computed per connected component of the graph would be tighter.

d) Grasp polytopes and offline cost: Our grasp polytope comes from one seed and a linearisation, and is valid near the seed only. More seeds from NLP sampling would cover other components of the grasp manifold. Merging overlapping pieces would allow longer strokes, but then placement must follow a chain of pieces. Growing IRIS regions from grasp samples produced many tiny polytopes, limited by the curvature of the manifold. A new obstacle layout requires a new decomposition, the dominant offline cost: in a separate run, IRIS-NP2 took 120 s for 25 regions at a 0.98 coverage target [CHECK:

different settings]. GPU implementations [41] reduce this cost, and new start and goal queries reuse the graph unchanged.

VIII. CONCLUSION

We posed multi-regrasp manipulation planning as a reachability problem over a convex decomposition of free space, and solved it with an A* search whose nodes pair polytopes with intervals of reachable object configurations. This transfers to manipulation an idea that recently matured in footstep planning. For objects with one coordinate coupled to the wrist, propagation is interval arithmetic and dominance is exact. The graph needs only LPs, and the search returns the minimum number of grasp actions relative to the decomposition, without fixing the number of regrasps in advance and without the numerical fragility of a mixed-integer formulation. On a 10-DOF Franka cap-turning task it plans in milliseconds where the MIQP fails. The same experiment shows that these guarantees are only as good as the decomposition near contact. We believe that certifying the switch sets, rather than covering free space, is the key requirement for planners of this kind. [TODO: one sentence on E1–E4 once run.]

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